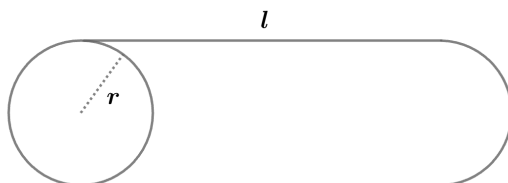


2018B F=ma Exam: Problem 25

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Theory:

$$A = 2\pi r l$$

We originally have

$$r = 1 \pm 0.1 \text{ cm}$$

$$l = 1.00 \pm 0.01 \text{ m}$$

Recall when two uncorrelated quantities are multiplied, the relative (fractional) uncertainties add in quadrature. Thus, the uncertainty in

Method 1 (change to $r = 1.00 \pm 0.01 \text{ cm}$):

$$\frac{\delta A}{A} = \sqrt{\left(\frac{\delta r}{r}\right)^2 + \left(\frac{\delta l}{l}\right)^2} = \sqrt{\left(\frac{0.01}{1.00}\right)^2 + \left(\frac{0.01}{1.00}\right)^2} = \sqrt{(1\%)^2 + (1\%)^2} = \mathbf{1.4\%}$$

Method 2 (change to $l = 1.00 \pm 0.001 \text{ m}$):

$$\frac{\delta A}{A} = \sqrt{\left(\frac{\delta r}{r}\right)^2 + \left(\frac{\delta l}{l}\right)^2} = \sqrt{\left(\frac{0.1}{1.0}\right)^2 + \left(\frac{0.001}{1.00}\right)^2} = \sqrt{(10\%)^2 + (0.1\%)^2} = \mathbf{10\%}$$

Method 3 (average 10 independent measurements): For a single measurement,

$$\frac{\delta A_0}{A} = \sqrt{\left(\frac{\delta r}{r}\right)^2 + \left(\frac{\delta l}{l}\right)^2} = \sqrt{\left(\frac{0.1}{1.0}\right)^2 + \left(\frac{0.01}{1.00}\right)^2} = \sqrt{(10\%)^2 + (1\%)^2} = 10\%$$

Recall when two uncorrelated quantities are added, their uncertainties add in quadrature. Averaging 10 measurements,

$$\delta A = \sqrt{\left(\frac{\delta A_0}{10}\right)^2 + \dots + \left(\frac{\delta A_0}{10}\right)^2} = \sqrt{10 \left(\frac{\delta A_0}{10}\right)^2} = \frac{\delta A_0}{\sqrt{10}}$$
$$\frac{\delta A}{A} = \frac{1}{\sqrt{10}} \frac{\delta A_0}{A} = \mathbf{3.2\%}$$

Thus, the answer is D.